

Department of Computer Science and Engineering - SVNIT, Surat
End Semester Examinations, December 2024
B.Tech. IV (Semester- VII)
Subject : CLOUD COMPUTING (CS403)

December 5, 2024

Time: 2pm to 5pm

Marks:50

Instructions:

1. Write your admission number/roll no. and other details clearly on the answer books while write your Admission No on the questions paper.
2. Be precise and clear in answering the question.
3. Begin each answer in a new page.

- Q.1
- a. Describe the key components of OpenStack and their roles. 03
 - b. Differentiate between time shared and space shared with respect to cloudsims. 02
 - c. Explain the concept of live migration in the context of virtual machines (VMs) and describe in detail how it enables the movement of a running VM from one physical host to another without interrupting its services or requiring downtime. 05
 - d. Explain trap and emulate technique to design virtual machine monitors. 03
 - e. Explain the concept address translation with paging. 02
 - f. Write pseudo-code to initialize hosts with specific CPU, memory, and bandwidth capacities in CloudSim. **OR** Provide pseudo-code to create a set of VMs with varying configurations and allocate them to a data center in CloudSim. 06
- Q.2
- a. Compare and contrast: Cloud Computing and Internet. **OR** Cloud Computing and SOA. 06
 - b. Explain types of hypervisors with diagram and example. 04
 - c. Explain types of full virtualization with diagram and example. 04
 - d. Explain the following term in detail: Risk profiling, VM Escape, OAuth, OpenID, SOAP 05
 - e. Which cloud deployment model is preferred for startups? Write its characteristics, advantages and disadvantages. 04
 - f. List out the resources that can be virtualized. Explain any two resource virtualizations in detail. 06
